Dynamic Social Optimum: Pigou Network

COMP3821

1 Definition and reduction to path assignments

The simulator uses linear edge costs $c_e(f) = a_e + b_e f$ and the total social cost

$$SC = \sum_t \sum_{e \in E} f_e(t)\, c_e\big(f_e(t)\big),$$

where $f_e(t)$ is the number of agents that traverse e during timestep t. The *dynamic social optimum* is the lowest value of this expression over every feasible way of routing the agents:

$$\boxed{SC_{\text{opt}} = \min_{\text{feasible routing decisions}} \sum_t \sum_{e \in E} f_e(t)\, c_e\big(f_e(t)\big)}$$

Feasible routings. The movement rules of the specification (Section 3.1) and the simulator fix what "feasible" means:

1. all N agents are released at $t = 0$ (none of these scenarios uses cascades);
2. at every timestep each unfinished agent traverses exactly one edge, so no agent can wait;
3. with `forward_only` (the default), an agent never moves to a node with smaller hop distance from its origin and never revisits a node.

Lemma 1 (Routing = path assignment)**.** *Let $\mathcal{P}_{od}$ be the simple paths from o to d that obey rule 3. Every feasible routing gives each agent a path $P_i \in \mathcal{P}_{o_i d_i}$, and agent i traverses the k-th edge of P_i at timestep $t = k - 1$. Every assignment of such paths can be realised. Therefore*

$$SC = \sum_{e \in E} \sum_{t \geq 0} X_{e,t}\, \big(a_e + b_e X_{e,t}\big), \qquad X_{e,t} = \#\{\, i : e \text{ is the } (t+1)\text{-th edge of } P_i \,\},$$

and SC_{opt} is the minimum of this over all path assignments.

Proof. Rule 2 means an agent's position at time t is the end of the t-th edge it has taken. Rule 3 means the edges it takes form a simple path in the allowed subgraph. So $f_e(t) = X_{e,t}$. Conversely, a central planner can tell each agent which next edge to commit to at each step. The outcome depends only on the paths, so every path assignment is achievable. □

So the dynamic problem is a *static* problem on the *time-expanded* resources (e, t). Agents on the same edge only slow each other down if they cross it in the same step.

Lemma 2 (When the dynamic and static optima coincide)**.** *Suppose every edge e sits at the same position $k(e)$ on every feasible path that contains it. Then $X_{e,t} = x_e\, \mathbf{1}[t = k(e) - 1]$, where x_e is the total number of agents using e. So $SC = \sum_e x_e\, c_e(x_e)$, which is the classical static social-optimum objective (Beckmann; Roughgarden) restricted to integer flows.*

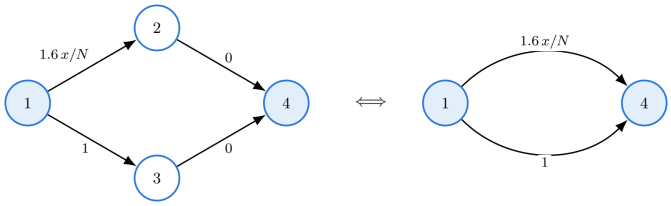
Integer versus continuous. Agents are indivisible, so SC_{opt} is an integer program. Relaxing the path counts to non-negative reals gives a convex quadratic program. Its minimum is a lower bound on SC_{opt} and equals the limit of SC_{opt}/N for large N. Both are reported below. Because c_e is linear, the cost one extra agent adds to resource (e,t) is

$$\Delta^+_{e,t} = (X+1)\,c_e(X+1) - X\,c_e(X) = a_e + b_e(2X+1), \qquad X = X_{e,t},$$

and in the continuous relaxation the marginal cost is $\frac{d}{dX}\big[Xc_e(X)\big] = a_e + 2b_eX$.

Scaling with N. Every small scenario sets $b_e = \beta_e/N$ with β_e fixed, so the per-agent optimum SC_{opt}/N is almost the same for $N = 20, 50, 80$.

2 Network



Edge	a_e	b_e	$\beta_e = b_eN$
$1 \to 2$	0	$1.6/N$	1.6
$2 \to 4$	0	0	0
$1 \to 3$	1	0	0
$3 \to 4$	0	0	0

Scenario	N	b_{12}
`pigou_N20`	20	0.08
`pigou_N50`	50	0.032
`pigou`	80	0.02

All agents travel $1 \to 4$. There are two feasible paths: $T = 1 \to 2 \to 4$ (top) and $B = 1 \to 3 \to 4$ (bottom).

3 Derivation

Timing. Both paths have two edges. Edges $1 \to 2$ and $1 \to 3$ are always crossed at $t = 0$, and $2 \to 4$ and $3 \to 4$ always at $t = 1$. Lemma 2 applies, so the dynamic and static optima coincide.

Objective. Let k agents take T and $N - k$ take B. Then $f_{12}(0) = k$ and $f_{13}(0) = N - k$, and the second-step edges cost nothing. So

$$SC(k) = k \cdot \frac{\beta k}{N} + (N-k)\cdot 1 = \frac{\beta}{N}k^2 - k + N, \qquad \beta = 1.6, \quad k \in \{0, 1, \ldots, N\}.$$

Continuous optimum. $SC(k)$ is a convex quadratic in k.

$$\frac{dSC}{dk} = \frac{2\beta k}{N} - 1 = 0 \implies k^* = \frac{N}{2\beta} = \frac{N}{3.2} = 0.3125\,N.$$

At k^* the marginal cost of the top route $(2\beta k/N)$ equals the bottom route's cost of 1, which is the textbook optimality condition. Substituting back:

$$SC(k^*) = \frac{\beta}{N}\cdot\frac{N^2}{4\beta^2} - \frac{N}{2\beta} + N = N\Big(1 - \frac{1}{4\beta}\Big) = N\Big(1 - \frac{1}{6.4}\Big) = 0.84375\,N.$$

Integer optimum. A quadratic is symmetric about its vertex, so over the integers it is minimised at the integer nearest to k^*:

$$N = 20:\ k^* = 6.25 \Rightarrow k = 6; \quad N = 50:\ k^* = 15.625 \Rightarrow k = 16; \quad N = 80:\ k^* = 25 \Rightarrow k = 25.$$

4 Result

N	Agents on T	Agents on B	SC_{opt}	SC_{opt}/N	Continuous bound $/N$
20	6	14	16.880	0.84400	0.84375
50	16	34	42.192	0.84384	0.84375
80	25	55	67.500	0.84375	0.84375

For example, at $N = 80$: $25 \cdot (0.02 \cdot 25) + 55 \cdot 1 = 12.5 + 55 = 67.5$.

$$\boxed{SC_{\text{opt}} = \tfrac{1.6}{N}k^2 + N - k \text{ with } k = \text{round}(N/3.2); \qquad SC_{\text{opt}}/N \to 0.84375}$$

Remark (Reference: Wardrop equilibrium). In the static selfish equilibrium, the top route is used until its cost reaches 1, at $k_{\text{UE}} = N/\beta = 0.625N$. Every agent then pays 1, so $SC_{\text{UE}} = N$ and the price of anarchy is $1/0.84375 = 32/27 \approx 1.185$.

Verification. The values were checked three independent ways, and all three agree exactly: (i) an exact successive-shortest-path solver for convex integer min-cost flow on the time-expanded network; (ii) a path-based solver for the continuous relaxation; (iii) rounding followed by single-agent improving moves.